

How Secure Is Your Base? A Recent Multi-Scanner Census of Linux Operating-System Images on Docker Hub

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Abstract. Every container starts from an operating-system (OS) base image, yet the security posture of these foundational images is usually measured with a single scanner and on samples that age quickly. We present a recent, multi-scanner census of the OS base images on Docker Hub: the **5,606** unique `amd64` images across the **21** repositories of Docker Hub’s Operating systems category, scanned with **14** independent open-source scanners spanning vulnerabilities, SBOM, misconfiguration, secrets and malware. Over the images processed to date we find that (i) vulnerability posture varies by an order of magnitude across distributions, with Debian and the RHEL family carrying the most critical and high findings and minimal or rolling-release distributions the fewest; (ii) staler images carry more vulnerabilities (Spearman $\rho = 0.16$, $p < 0.001$; bucketed means rising from ≈ 255 to ≈ 692); (iii) the four package-vulnerability engines barely agree (pairwise Jaccard ≤ 0.35), with OSV-Scanner and Clair nearly disjoint from the rest; and (iv) about one in five historically-published OS images can no longer be pulled by a modern Docker because they use a legacy manifest schema. We release the pipeline and the consolidated multi-scanner dataset, to our knowledge the first OS-base-focused one, on acceptance.

1. Introduction

Containers are now a default unit of software delivery: Docker reports more than 20 million developers using its tools and, in its 2025 survey, 92% of IT professionals using containers [Docker, Inc. 2025], while the Cloud Native Computing Foundation finds most organisations running the majority of their applications in containers [Cloud Native Computing Foundation 2025]. Underneath them sits Docker Hub, the largest public registry, with on the order of 8.3 million image repositories and hundreds of billions of cumulative pulls [Docker, Inc. 2024]. Every one of those images ultimately begins with a single line, FROM an operating-system (OS) base image, and a handful of bases are the common root of nearly all of them: `alpine`, `ubuntu`, `debian` and `busybox` each record more than a billion pulls, and Debian, Alpine and Ubuntu dominate the base-image choice of the ecosystem [Anchore 2024]. Because a flaw in a base image is inherited, unchanged, by every image built on it [Shu et al. 2017], the security posture of these few OS bases is a property the whole ecosystem inherits.

That inherited posture is poor and consequential. Software-supply-chain attacks are projected to cost USD 60 billion in 2025 [Cybersecurity Ventures 2023] and Gartner predicted that 45% of organisations would suffer such an attack by 2025 [Gartner, Inc. 2021]; meanwhile popular Docker Hub images carry hundreds of known vulnerabilities on average, many of them years old [NetRise 2024], against a backdrop of record CVE disclosure (46,407 CVEs in 2025, about 127 per day) [NIST National Vulnerability Database 2025]. Worse, some of the most-pulled OS bases are no longer maintained (every official `centos` tag reached end-of-life in June 2024) yet remain in heavy use.

Despite this central role, the security posture of OS base images has usually been characterised with a *single* scanner and on samples that age quickly: the large-scale Docker Hub measurements have typically used one detector [Shu et al. 2017, Zerouali et al. 2019, Liu et al. 2020, Shi et al. 2025], and the studies that *do* compare scanners have done so on focused samples [Kaur et al. 2021, Mills et al. 2023]. No recent work measures the Linux OS base images specifically, with a heterogeneous multi-scanner battery, as a current census. This paper fills that gap.

We make the following contributions:

- A **recent multi-scanner census** of OS base images on Docker Hub: 14 scanners over 5,606 unique `amd64` images of the 21 *Operating systems* repositories (Section 3).
- A **per-distribution security posture** and a measured **staleness-vulnerability** relationship (Sections 4.1, 4.2).
- A **four-engine divergence** measurement on the OS-package layer, including Clair, showing the chosen scanner dominates the reported posture (Section 4.3).
- A supply-chain **longevity finding**: a large share of historically-pulled OS images are no longer installable by a modern Docker (legacy manifest schema), and end-of-life bases remain heavily pulled.
- A **released artifact and dataset**. We release the scan pipeline and the consolidated multi-scanner dataset, the per-image reports of all 14 scanners over the OS census; to our knowledge it is the first OS-base-focused multi-scanner dataset, letting the measurement be audited, reproduced and extended (released on acceptance).

2. Related Work

A rich line of work has charted the security of Docker Hub. [Shu et al. 2017] introduced the first large-scale study, scanning 356,218 images and showing that vulnerabilities propagate from a base image to everything built on it. [Liu et al. 2020] extended the scale to 2.2 million images; [Zerouali et al. 2019] introduced the notion of *technical lag* on 7,380 Debian-based images, later generalised to a multi-dimensional analysis over 140,000 images [Zerouali et al. 2021]. [Wist et al. 2021] analysed 2,500 images across image classes, [Ibrahim et al. 2020] examined how images for the same system differ across base distributions, and the recent [Shi et al. 2025] ranked influential images by pull count at the scale of the full registry. Most relevant to us, [Haque and Babar 2022] characterised the exploitability and impact of vulnerabilities in 261 official base images and reported a result we revisit in Section 4.5: highly-exploitable vulnerabilities are *more* frequent in minimal images such as Alpine, while larger distributions carry the higher-impact ones. These studies established the prevalence and propagation of vulnerabilities in the ecosystem; each draws on a single vulnerability detector, and our multi-scanner design complements them by also quantifying how much the measured posture depends on the chosen scanner.

A second line studies how much a scan result depends on the tool used. [Imtiaz et al. 2021] compared nine software-composition-analysis tools on the same projects and found counts spanning an order of magnitude, and [Javed and Toor 2021a, Javed and Toor 2021b] report comparable variation among container scanners. For containers, careful side-by-side comparisons have been conducted on focused samples: [Kaur et al. 2021] use four scanners over 44 images, [Mills et al. 2023] six over 380, and the recent

[Churakova et al. 2026] evaluate seven scanner and VEX tools on 48 images, finding counts from 191 to 18,680 and a best-pair (Trivy and Grype) agreement of about 69%; closest to us in subject, [Boles et al. 2024] compared Trivy and Grype on 927 Docker Official Linux images and found the two agreeing on the vulnerability count for only 15% of them. On the OS-package layer specifically, [Bufalino et al. 2025] report that current scanners are largely incompatible because of package-identifier mismatches, and [Zhou et al. 2026, Rosso et al. 2026] document high false-positive rates and silent tool failures in SBOM-based scanning. Our work is complementary, bringing this multi-scanner lens to the OS-base layer at census scale, adding Clair, and relating the divergence to image age and distribution.

Building on this body of work, we are not aware of a recent study that measures the Linux OS base images of Docker Hub specifically with a heterogeneous multi-scanner battery as a current census, nor one that quantifies the prevalence and popularity of *end-of-life* OS images that remain in heavy use. This paper addresses both.

Table 1. Prior OS/base-image and container-scanner measurements.

Study	Year	Images	Tools	Dims	Source	OS	X-sc.
[Shu et al. 2017]	2017	356,218	1	V	Hub crawl	✗	✗
[Zerouali et al. 2019]	2019	7,380	1 [‡]	V	Hub, Debian	✓	✗
[Liu et al. 2020]	2020	2.2 M*	1	V	Hub crawl	✗	✗
[Ibrahim et al. 2020]	2020	936	1	V	Hub, per-sys	✗	✗
[Kaur et al. 2021]	2021	44	4	V	curated	✗	✓
[Zerouali et al. 2021]	2021	140,498	1 [‡]	V	Hub, Debian	✓	✗
[Haque and Babar 2022]	2022	261	1	V	Hub, official	✓	✗
[Dahlmanns et al. 2023]	2023	337,171	1	S	Hub crawl	✗	✗
[Malhotra et al. 2023]	2023	n/r	4	V	Hub off./ver.	✗	✓
[Mills et al. 2023]	2023	380	6	V	Hub categories	✗	✓
[Boles et al. 2024]	2024	927	2	V	Hub official	✓	✓
[Kawaguchi et al. 2024]	2024	3,514	2	B	Hub popular	✗	✓
[Bufalino et al. 2025]	2025	20 [†]	7	V	Hub Deb/Alp	✓	✓
[Shi et al. 2025]	2025	33,952*	1	V,S,C,M	Hub pull-rank	✗	✗
[Churakova et al. 2026]	2026	48	7	V	Hub curated	✗	✓
This work	2026	5,606	14	V,B,S,C,M	Hub category	✓	✓

Dims: V vuln, B SBOM, S secrets, C config, M malware. X-sc.: cross-scanner. * vulnerability scan on a subset; † top-20 Debian/Alpine image families; ‡ Debian Security Tracker (metadata), not a scanner.

3. Methodology

We take as our corpus the operating-system base images that Docker Hub itself lists under its *Operating systems* category: 22 repositories (e.g. alpine, ubuntu, debian, centos, busybox, amazonlinux, fedora, oraclelinux, rockylinux, almalinux, photon, opensuse, archlinux, kali), one of which (clefos) publishes no linux/amd64 image and drops out. Using the Docker Hub API we enumerated *all* tags of these repositories, **6,328** tags in total, of which **6,138** carry an amd64/linux image. We note in passing that the “10K+” tag badge Docker Hub

shows for `ubuntu` and `alpine` counts `tag` $\times$ `architecture` $\times$ `attestation manifests`, not distinct tags: `ubuntu` has 725 distinct tags and `debian` 2,884. Because many tags point to a byte-identical image (e.g. `24.04`, `noble` and `latest`), we deduplicate by `amd64` content digest, leaving **5,606** unique images, the unit of measurement throughout.

An OS base image is a static root filesystem with no application source code and no running service, so we select the static-analysis axes that apply to such an artifact and run 14 open-source scanners.¹ *SBOM*: Syft and cdxgen. *Package vulnerabilities (SCA)*: Trivy, Gripe, OSV-Scanner and Clair. *Image hardening*: Dockle and Checkov. *Secrets*: TruffleHog, Gitleaks, detect-secrets and Whispers. *Malware/IOC*: ClamAV and YARA-Hunter. We deliberately exclude the SAST, dynamic (DAST) and network axes, which require application source or a running service that a base image does not provide; we make this selection explicit so the battery matches the artifact under study.

Each image is pulled *pinned* by its `amd64` digest, exported once to a local tar archive, and analysed by all 14 scanners reading that archive; the image is removed afterwards to bound disk. Every scanner runs in its own Docker container and a per-scanner adapter normalises the raw output to a common `Finding` schema; raw outputs are kept verbatim. To obtain a balanced partial dataset at any stopping point, the scan queue is ordered *round-robin* across the 21 repositories, and within each repository it alternates newest $\leftrightarrow$ oldest by last-push date, so the full age range of every distribution is covered early.

4. Results

The census is still running; the figures and numbers below are computed over the **1,927** images scanned so far and are regenerated as it completes.

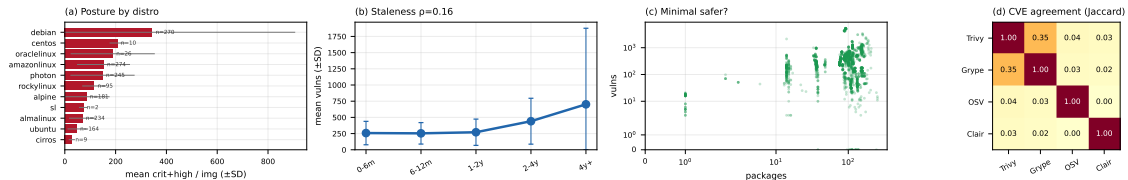


Figure 1. Posture by distribution (a), staleness (b), packages vs. vulnerabilities (c), SCA-engine Jaccard (d). Bars: ± 1 SD.

4.1. RQ1: Security posture by distribution

Critical- and high-severity vulnerability findings vary by an order of magnitude across distributions (Figure 1(a)). Debian carries the most (mean ≈ 344 critical+high per image, $n = 270$), followed by the RHEL family (CentOS ≈ 206 , OracleLinux ≈ 189 , AmazonLinux ≈ 152 , RockyLinux ≈ 115); Alpine sits lower (≈ 84), and Arch, ALT, Kali, Mageia and openSUSE Tumbleweed report few or none. Within-distribution variance is large (for Debian the standard deviation, ≈ 564 , exceeds the mean), so these are population tendencies rather than tight estimates, and the smaller repositories (for example CentOS, $n = 10$) should be read with care. Package count does not track this ordering: Mageia and Fedora carry many packages yet few critical or high findings, a point we return to in Section 4.5.

¹Each scanner runs from a version-pinned container image; the exact versions are recorded in the released configuration.

4.2. RQ2: Staleness vs. vulnerability

Vulnerability count tends to rise with image age (Figure 1(b)). Images rebuilt within the last six months carry a mean of ≈ 255 total vulnerability findings, climbing through ≈ 273 (one to two years) and ≈ 457 (two to four years) to ≈ 692 for images not rebuilt in over four years. At the image level the rank correlation between age and vulnerability count is positive and significant, though modest (Spearman $\rho = 0.16$, $p < 0.001$, $n = 1,927$), reflecting wide within-bucket variance; the bucketed means themselves increase monotonically. This reproduces the technical-lag effect of [Zerouali et al. 2019] on the OS-base subset and across 21 distributions at once.

4.3. RQ3: Inter-scanner divergence on OS packages

The four package-vulnerability engines barely agree (Figure 1). We compare at the (image, CVE) level because the engines name packages differently (Clair reports source packages, Trivy and Gype binary packages), so a package-level key would understate agreement. Even so, the best-agreeing pair is Trivy and Gype, at a Jaccard of just **0.35**; every other pair is below 0.05, and OSV-Scanner and Clair are near-disjoint from the rest ($\text{OSV} \cap \text{Clair} = 0$). The engines differ sharply in scope: Gype and Trivy report by far the most image-CVE pairs (196k and 148k), OSV-Scanner targets language-ecosystem advisories (9k) and Clair maps only specific distributions to its feeds (8k, and zero on Rocky and AlmaLinux, which it does not map to the RHEL feeds). The reported security posture of an OS image is thus, to a large degree, a property of the chosen scanner, echoing [Churakova et al. 2026] at far larger scale and on the OS layer specifically.

4.4. RQ4: End-of-life and un-pullable images

Two longevity hazards surface. First, about **one in five** of the digest-pinned OS images attempted so far fail to pull on a current Docker Engine with `manifest schema unsupported`: they are old images in the deprecated Docker image-manifest schema 1, whose support modern Docker removed. The digests still exist on Docker Hub, but the images are no longer installable, and the rate varies sharply by distribution (Figure 2): over half of the `centos` and `busybox` digests and about a third of `ubuntu` are affected, against under 3% for `debian`, `archlinux` and `rockylinux`. Second, end-of-life distributions remain in heavy use: every `centos` tag has been unmaintained since June 2024, yet the repository still records over a billion pulls. To our knowledge neither the prevalence of legacy-schema OS images nor the popularity of EOL bases has been measured before.

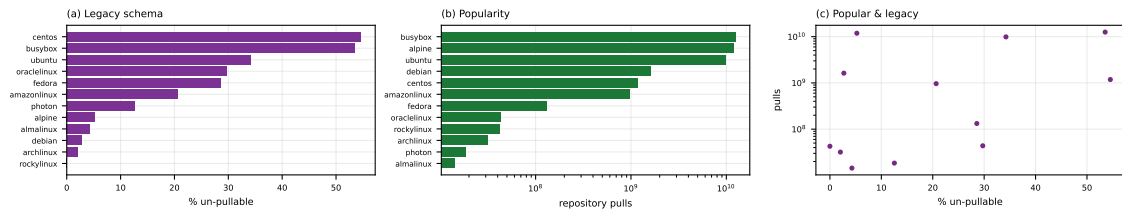


Figure 2. (a) un-pullable rate by distribution; (b) repository pulls; (c) rate vs. pulls.

4.5. RQ5: Is a minimal image safer?

The intuition that a smaller image is a safer one does not hold cleanly (Figure 1(c)). The relationship between package count and vulnerability count is not monotonic: BusyBox,

with a single package, carries only a handful of findings, but Mageia, with around 250 packages, carries about a dozen, while Debian carries hundreds of critical and high findings with fewer packages than Mageia. This refines [Haque and Babar 2022]: on our multi-scanner data the vulnerability count is governed more by a distribution’s advisory coverage and patch cadence than by image size.

4.6. Beyond vulnerabilities: secrets, misconfiguration, malware

The battery also covers four non-vulnerability axes. *Misconfiguration*: Dockle and Checkov flag at least one CIS-style issue in essentially every image (missing HEALTHCHECK, content-trust disabled, running as `root`); these are structural properties, so the near-universal rate is robust. *Secrets*: TruffleHog and Gitleaks raise at least one detection in **72%** of images, but these are raw detector hits: consistent with [Dahlmanns et al. 2023], who validate secrets in only 8.5% of images, and with Dr. Docker [Shi et al. 2025], which filters 99.3% of its raw hits as invalid, the validated rate is far lower, and active validation is future work. *Malware/IOC*: ClamAV and YARA-Hunter raise IOC matches on many images, but manual inspection shows these are dominated by signatures matching benign system binaries; we treat them as triage candidates rather than confirmed malware (Dr. Docker confirmed only 24 malicious images in 33,952). These axes are secondary to our vulnerability focus, but they show the same lesson as RQ3: raw single-tool counts overstate, and must be read with their false-positive rate.

4.7. Reproducing prior analyses

Our four questions reproduce, on the OS-base subset specifically, concrete analyses from prior work (Table 2); the direction holds in every case, and the multi-scanner setting lets us refine the minimal-image and scanner-agreement results that single-tool studies could not.

Table 2. Prior analyses reproduced on our OS-base corpus.

Study	Metric reproduced	Reported	Ours
[Shu et al. 2017]	images with a high-severity vuln	>80%	97%*
[Zerouali et al. 2019]	vulnerabilities vs. image age (Debian)	rises with age	$\approx 255 \rightarrow 692$ ($\rho=0.16$)
[Ibrahim et al. 2020]	spread across base distributions	varies	10 $\times$ range
[Wist et al. 2021]	where severe vulns concentrate	language ecosystems	OS-package layer
[Haque and Babar 2022]	is a minimal image safer?	no	no (non-monotonic)
[Dahlmanns et al. 2023]	images carrying secrets	8.5% (valid.)	72% (raw)
[Mills et al. 2023]	vulnerabilities vs. time; OS effect	rise; Debian $\gg$	rise; Debian top
[Boles et al. 2024]	scanner divergence on OS images	agree 15%	Jaccard 0.35
[Shi et al. 2025]	known-vulnerability prevalence	93.7%	98.7%

* critical or high (Shu reports high-severity); ρ : Spearman.

4.8. Threats to validity

Several threats bound these results. *Coverage*: the census is still running, so the numbers reflect the 1,927 images processed to date; the round-robin order keeps every distribution represented, but the smaller repositories carry wide uncertainty and their means may shift as the run completes. *Scanner error*: we report raw scanner output and do not adjudicate false positives or negatives, and the divergence we measure (RQ3) itself implies that no single engine is ground truth; severity, moreover, rests on CVSS scores that can overstate operational risk without VEX or reachability information [Zhou et al. 2026]. *Architecture*: we scan `linux/amd64` only, so the findings may not transfer to `arm64`. *Temporal drift*: a tag can be re-pushed during the census, so each image is characterised as of its scan time. *Generality*: the corpus is Docker Hub’s *Operating systems* category, so other registries (Quay, GHCR) and non-base images are out of scope. The released dataset lets others re-run the analysis under different choices.

5. Conclusion

We presented a recent, multi-scanner census of the operating-system base images of Docker Hub, the bases on which nearly every container is built, scanned with 14 independent open-source tools rather than one. Across the images measured so far, vulnerability posture varies by an order of magnitude across distributions and rises steadily with image age; the four package-vulnerability engines agree on almost nothing, so the posture a single tool reports is in large part a property of that tool; a smaller image is not reliably a safer one; and about a fifth of historically-published OS images can no longer be installed by a modern Docker, while end-of-life bases such as CentOS remain heavily pulled. We release the pipeline and the consolidated dataset² so the measurement can be reproduced and kept current.

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²The scan pipeline, analysis code and the consolidated multi-scanner dataset will be released on acceptance.

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